

Program1

Build a Docker container from a custom Docker file

Step1: create the following files inside a folder [app.py](#), Dockerfile, requirements.txt

Pg1

```
|  
|-----app.py  
|-----Dockerfile  
|-----requirements.txt
```

[app.py](#)

```
from flask import Flask  
  
app=Flask(__name__)  
@app.route("/")  
def hello():  
    return "Hello world"  
  
if __name__=="__main__":  
    app.run(host="0.0.0.0",port=5000)
```

Dockerfile

```
FROM python:3.9-slim  
  
WORKDIR /app  
  
COPY requirements.txt .  
RUN pip install --no-cache-dir -r requirements.  
  
COPY . .  
  
ENV FLASK_APP=app.py  
ENV FLASK_RUN_HOST=0.0.0.0  
ENV FLASK_RUN_PORT=5000  
  
EXPOSE 5000  
  
CMD ["flask","run"]
```

Step2: Build and run Dockerfile

```

lrv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg1$ docker build -t program1 .
[+] Building 10.6s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 270B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 2.6s
=> [auth] library/python:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f1 0.0s
=> [internal] load build context                                   0.0s
=> => transferring context: 532B                                       0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> [3/5] COPY requirements.txt .                                    0.0s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt        7.7s
=> [5/5] COPY . .                                                  0.0s
=> exporting to image                                              0.2s
=> => exporting layers                                              0.2s
=> => writing image sha256:3c469e87c2134c559a0c408b7c6946f6d9d2efd594e79fc64eca5219390 0.0s
=> => naming to docker.io/library/program1                        0.0s
lrv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg1$ docker run -p 5000:5000 program1
* Serving Flask app 'app.py'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a product
ion WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit

```

Output:

